Brooke Ballhaus

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AI engineer focused on thoughtful systems design: machines take the high-volume, repetitive work so people can focus on the judgment and expertise only they bring.

EXPERIENCE

General Dynamics Information Technology | AI/ML Technical Intern — Falls Church, VA | 06/2024 - Current

- Architected and built PAICUS, a human-in-the-loop AI platform prototype for CMS's Center for Consumer Information and Insurance Oversight (CCIIO): AI reads and analyzes complex healthcare policy while analysts keep every decision
- Proposed, architected, and built an end-to-end fraud, waste, and abuse detection pipeline over 50 million dental claims on AWS (BioBERT embeddings, K-Means and autoencoder anomaly detection), with Claude explaining each flagged claim in plain English so investigators spend their time on judgment calls, not data triage
- Created and taught "AI: Beyond the Basics," an internal training for 60+ attendees, recorded for company-wide viewing, that equips program managers to get real value from AI tools
- Chosen as Day 1 Keynote Speaker at GDIT's FedCiv Pursuit Palooza 2025, pitching "Unconstrained Thinking" in AI solution design to Federal Civilian business leaders
- Built and presented LLM solutions across 6 projects for the Health Analytics Lighthouse and Luna AI Digital Accelerator, including a multi-model system that automates verification of handwritten healthcare forms; ran SQL spike analysis on federal claims data and delivered findings to the client in Tableau
- Selected from 117 interns to present to the GDIT and General Dynamics Executive Leadership Teams at the August Operations Review; won the end-of-summer award for excellence

NASA Goddard Space Flight Center | Software/Systems Engineering Intern (via Peraton) — Greenbelt, MD | 06/2023 - 05/2024

- Built automated report generation for modem test suites on NASA's Near Space Network (Python backend and GUI), later extended to generate testing procedures

Stanford University | CS107 Teaching Assistant — Stanford, CA | 09/2026 - Current

- Lead weekly lab sections for Stanford's systems programming course, teaching C, memory, and debugging with gdb and Valgrind to 30 students

Stanford Equestrian Team | President — Stanford, CA | 05/2023 - 06/2026

- Led a 40-member team with 35 horses and a $200,000+ annual budget; ran seven home competitions per year and rebuilt the tryout process for ~100 applicants annually

Stanford Society of Women Engineers | Outreach Intern — Stanford, CA | 09/2022 - 06/2026

- Planned six events facilitating 20 mentorships between Stanford and Bay Area high school students interested in STEM

EDUCATION

Stanford University | Master of Science, Computer Science (Artificial Intelligence) — Stanford, CA | Class of 2027

- Coursework: NLP with Deep Learning, Deep RL, Trustworthy ML, Decision Making Under Uncertainty, Computer Vision, Building LLM Products

Stanford University | Bachelor of Science, Computer Science (Artificial Intelligence) — Stanford, CA | Class of 2026

- Activities: Stanford Equestrian Athlete (competed at Zones and Nationals); Ram's Head Theater Actor (three musicals); Alpha Phi Sisterhood Director; Stanford Polo Club Competing Member

PROJECTS

Civitas – AI procurement intelligence for small government contractors. Led a 5-person team and wrote the majority of 640+ commits: scrapers monitor 62 California procurement sites and semantic matching (Postgres/pgvector) surfaces the right RFPs, so contractors spend their time winning work instead of hunting for it

crewAI (open source, merged) – Contributed an offline evaluation module to crewAI, a popular multi-agent framework, scoring agent routing and guardrail decisions; 46 tests, strict type checking, mutation-tested

Prose.ai (in progress) – Website builder for non-coders where a structured English spec, not a chat log, is the source code: each sentence compiles through Claude into its own tested module, so one change never breaks unrelated parts

Rooted (CS177, in progress) – AI decision partner for startup product managers that pulls context from the team's tools, helps weigh options, and keeps a searchable record of what was decided and why; the PM still makes the call

When Can We Trust AI Explanations? – Mechanistic audit of chain-of-thought faithfulness in Llama-3-8B (logit lens, activation patching); found it commits to its answer in the last ~3 of 32 layers whether or not it reasons first

Saipiens – AI tutor for K-12 that coaches students through their reasoning instead of handing out answers; 212+ automated tests

SKILLS

Languages & ML: Python, TypeScript, SQL, C; PyTorch, Hugging Face, TransformerLens, LangChain; Claude/Gemini/Llama APIs; RAG, LLM evals

Systems & Cloud: System design, testing, security hardening; AWS (Lambda, S3, DynamoDB, RDS, Cognito), Next.js, React, Docker